

From the harmonic toy to the sech^2 disk: A complete derivation of the σ_w/σ_z mismatch asymmetry

1 The question

We have a 1D vertical phase-space simulation. Stars are sampled at $t = 0$ from a Gaussian

$$f_0(z_0, w_0) = \frac{1}{2\pi\sigma_z\sigma_w} \exp\left[-\frac{z_0^2}{2\sigma_z^2} - \frac{w_0^2}{2\sigma_w^2}\right] \quad (1)$$

and integrated forward to a time t in a vertical potential $\Phi(z; h)$, where h is the dark-matter scale-height parameter we want to fit. We observe $\{(z_{\text{obs}}^{(i)}, w_{\text{obs}}^{(i)})\}_{i=1}^N$ at t and recover h by maximum likelihood: for a trial h , back-integrate each star to $t = 0$, then evaluate the log-prior of the back-integrated cloud under *assumed* sigmas $(\sigma_{z,\text{fit}}, \sigma_{w,\text{fit}})$.

The empirical observation we want to explain:

- Shrink $\sigma_{w,\text{fit}}$ by a factor r ($r = 10$ or 100). The fitted h runs away from the truth.
- Shrink $\sigma_{z,\text{fit}}$ by the same factor r . The fit barely moves.

We will derive the reason in two steps. First, the harmonic potential, where every quantity is in closed form. Then the real ($\text{sech}^2 + \text{DM}$) potential, where the same structure holds but the algebra has to be done as orbit averages.

2 Harmonic potential: full closed-form derivation

2.1 The potential, the Hamiltonian, and the equations of motion

Consider a single particle of unit mass in

$$\Phi(z) = \frac{1}{2}\nu^2 z^2, \quad H(z, w) = \frac{1}{2}w^2 + \frac{1}{2}\nu^2 z^2. \quad (2)$$

Hamilton's equations are

$$\dot{z} = \frac{\partial H}{\partial w} = w, \quad \dot{w} = -\frac{\partial H}{\partial z} = -\nu^2 z. \quad (3)$$

The frequency ν is the analog of the dark-matter parameter h in this toy: it is the parameter the fitter tries to recover.

2.2 The flow

The general solution of (3) starting from (z_0, w_0) at $t = 0$ is

$$\boxed{z(t) = z_0 \cos(\nu t) + \frac{w_0}{\nu} \sin(\nu t), \quad w(t) = -\nu z_0 \sin(\nu t) + w_0 \cos(\nu t).} \quad (4)$$

This is a rotation by angle $\theta = \nu t$ in the rescaled coordinates $(z, w/\nu)$ — symplectic and area-preserving.

2.3 The back-integration map

Inverting (4) at trial frequency ν , given observations $(z_{\text{obs}}, w_{\text{obs}})$ at time t :

$$\boxed{z_0(\nu) = z_{\text{obs}} \cos(\nu t) - \frac{w_{\text{obs}}}{\nu} \sin(\nu t), \quad w_0(\nu) = \nu z_{\text{obs}} \sin(\nu t) + w_{\text{obs}} \cos(\nu t).} \quad (5)$$

Throughout the rest of this section, $(z_{\text{obs}}, w_{\text{obs}})$ are *fixed by the data*, generated by (4) at the true frequency ν_t , while ν is the trial frequency the fitter is varying.

2.4 The truth flow inserted into the back-integration

What we want and why. The back-map (5) gives $z_0(\nu)$ and $w_0(\nu)$ as linear functions of the *observations* $z_{\text{obs}}, w_{\text{obs}}$. That's not yet useful for computing variances, because the observations are themselves random (they depend on the unknown truth z_0, w_0). We want $z_0(\nu)$ and $w_0(\nu)$ expressed in terms of the *truth* z_0, w_0 — the variables whose statistics we know (Gaussian with σ_z, σ_w).

The idea: chain two linear maps.

- The back-map (eq. (5)) is linear in $(z_{\text{obs}}, w_{\text{obs}})$.
- The forward truth-flow (eq. (4) at ν_t) makes $(z_{\text{obs}}, w_{\text{obs}})$ linear in (z_0, w_0) .
- Substituting the second into the first, $(z_0(\nu), w_0(\nu))$ becomes linear in (z_0, w_0) .

The coefficients of this composite linear map are exactly A, B, C, D . Concretely: $A(\nu)$ and $C(\nu)$ are the coefficients of z_0 and w_0 in $z_0(\nu)$; $B(\nu)$ and $D(\nu)$ are the same for $w_0(\nu)$. No tricks — substitute and collect.

Explicit substitution. Substituting the truth flow

$$z_{\text{obs}} = z_0 \cos \theta_t + \frac{w_0}{\nu_t} \sin \theta_t, \quad w_{\text{obs}} = -\nu_t z_0 \sin \theta_t + w_0 \cos \theta_t, \quad \theta_t \equiv \nu_t t, \quad (6)$$

into (5) (and writing $\theta \equiv \nu t$) gives

$$z_0(\nu) = A(\nu) z_0 + C(\nu) w_0, \quad w_0(\nu) = B(\nu) z_0 + D(\nu) w_0, \quad (7)$$

with

$$A(\nu) = \cos \theta_t \cos \theta + \frac{\nu_t}{\nu} \sin \theta_t \sin \theta, \quad (8)$$

$$C(\nu) = \frac{1}{\nu_t} \sin \theta_t \cos \theta - \frac{1}{\nu} \cos \theta_t \sin \theta, \quad (9)$$

$$B(\nu) = \nu \cos \theta_t \sin \theta - \nu_t \sin \theta_t \cos \theta, \quad (10)$$

$$D(\nu) = \frac{\nu}{\nu_t} \sin \theta_t \sin \theta + \cos \theta_t \cos \theta. \quad (11)$$

Sanity check at $\nu = \nu_t$: $A = D = 1$ and $B = C = 0$, so $z_0(\nu_t) = z_0$ and $w_0(\nu_t) = w_0$. The back-integration recovers the truth.

2.5 The variances of the back-integrated cloud

Because $\langle z_0 w_0 \rangle = 0$ under the prior (1):

$$\boxed{V_z(\nu) \equiv \langle z_0(\nu)^2 \rangle = \sigma_z^2 A(\nu)^2 + \sigma_w^2 C(\nu)^2,} \quad (12)$$

$$\boxed{V_w(\nu) \equiv \langle w_0(\nu)^2 \rangle = \sigma_z^2 B(\nu)^2 + \sigma_w^2 D(\nu)^2.} \quad (13)$$

At truth: $V_z(\nu_t) = \sigma_z^2$, $V_w(\nu_t) = \sigma_w^2$. The fit at ν_t recovers the correct sigmas.

2.6 The likelihood

Up to ν -independent constants, the per-star negative log-likelihood the fitter optimises is

$$\text{NLL}(\nu) = \frac{V_z(\nu)}{2\sigma_{z,\text{fit}}^2} + \frac{V_w(\nu)}{2\sigma_{w,\text{fit}}^2}. \quad (14)$$

Differentiating in ν :

$$\boxed{\text{NLL}'(\nu) = 0 \quad \Longleftrightarrow \quad \frac{V_z'(\nu)}{\sigma_{z,\text{fit}}^2} + \frac{V_w'(\nu)}{\sigma_{w,\text{fit}}^2} = 0.} \quad (15)$$

This is the equation the fitter solves for $\nu_\star$ (the MLE).

2.7 First derivatives at $\nu = \nu_t$

Differentiate A, B, C, D once with respect to ν and evaluate at $\nu = \nu_t$. Keeping all terms (not yet asymptotic in t):

$$A'(\nu_t) = -\frac{\sin^2 \theta_t}{\nu_t}, \quad (16)$$

$$C'(\nu_t) = -\frac{t}{\nu_t} + \frac{\sin 2\theta_t}{2\nu_t^2}, \quad (17)$$

$$B'(\nu_t) = \nu_t t + \frac{\sin 2\theta_t}{2}, \quad (18)$$

$$D'(\nu_t) = +\frac{\sin^2 \theta_t}{\nu_t}. \quad (19)$$

Then, using $A(\nu_t) = D(\nu_t) = 1$ and $B(\nu_t) = C(\nu_t) = 0$:

$$V_z'(\nu_t) = 2\sigma_z^2 A(\nu_t) A'(\nu_t) + 2\sigma_w^2 C(\nu_t) C'(\nu_t) = -\frac{2\sigma_z^2 \sin^2 \theta_t}{\nu_t}, \quad (20)$$

$$V_w'(\nu_t) = 2\sigma_z^2 B(\nu_t) B'(\nu_t) + 2\sigma_w^2 D(\nu_t) D'(\nu_t) = +\frac{2\sigma_w^2 \sin^2 \theta_t}{\nu_t}. \quad (21)$$

$$\boxed{V_z'(\nu_t) = -\frac{2\sigma_z^2 \sin^2 \theta_t}{\nu_t}, \quad V_w'(\nu_t) = +\frac{2\sigma_w^2 \sin^2 \theta_t}{\nu_t}.} \quad (22)$$

Truth-balance check. With $\sigma_{z,\text{fit}} = \sigma_z$ and $\sigma_{w,\text{fit}} = \sigma_w$, (15) at ν_t reads

$$-\frac{2\sin^2\theta_t}{\nu_t} + \frac{2\sin^2\theta_t}{\nu_t} = 0, \quad (23)$$

so the MLE sits exactly at ν_t . The two variance gradients pull in opposite directions, and at truth sigmas they cancel exactly.

2.8 Second derivatives at $\nu = \nu_t$ — the key asymmetry

We need

$$V_z''(\nu_t) = 2\sigma_z^2[(A')^2 + AA'']|_{\nu_t} + 2\sigma_w^2[(C')^2 + CC'']|_{\nu_t}, \quad (24)$$

$$V_w''(\nu_t) = 2\sigma_z^2[(B')^2 + BB'']|_{\nu_t} + 2\sigma_w^2[(D')^2 + DD'']|_{\nu_t}. \quad (25)$$

We need A'', B'', C'', D'' at ν_t . Direct differentiation, keeping only the leading t^2 terms (large- t , many-orbits regime), gives

$$A''(\nu_t) \simeq -t^2, \quad D''(\nu_t) \simeq -t^2, \quad (26)$$

and the leading large- t behaviours

$$C'(\nu_t) \simeq -\frac{t}{\nu_t} \Rightarrow [C'(\nu_t)]^2 \simeq \frac{t^2}{\nu_t^2}, \quad B'(\nu_t) \simeq \nu_t t \Rightarrow [B'(\nu_t)]^2 \simeq \nu_t^2 t^2. \quad (27)$$

Plugging in, with $A = D = 1$ and $B = C = 0$ at ν_t , only the $AA'', DD'', (B')^2, (C')^2$ terms survive at order t^2 :

$$V_z''(\nu_t) \simeq 2\sigma_z^2(-t^2) + 2\sigma_w^2 \frac{t^2}{\nu_t^2} = 2t^2 \left(\frac{\sigma_w^2}{\nu_t^2} - \sigma_z^2 \right), \quad (28)$$

$$V_w''(\nu_t) \simeq 2\sigma_z^2 \nu_t^2 t^2 + 2\sigma_w^2(-t^2) = 2t^2(\sigma_z^2 \nu_t^2 - \sigma_w^2). \quad (29)$$

2.9 The dimensionless ratio R

Define

$$R \equiv \frac{\sigma_w}{\nu_t \sigma_z} = \frac{\text{velocity-dispersion-induced amplitude}}{\text{position-dispersion amplitude}}. \quad (30)$$

In the simulation $\sigma_w \sim 20$ km/s, $\sigma_z \sim 10$ pc, $\nu_t \sim 0.04$ Myr $^{-1}$ (after unit-conversion), so $R \sim 50$. We will assume $R \gg 1$.

The second derivatives become

$$V_z''(\nu_t) = 2\sigma_z^2 t^2 (R^2 - 1) \xrightarrow{R \gg 1} + 2\sigma_z^2 t^2 R^2, \quad (31)$$

$$V_w''(\nu_t) = 2\sigma_w^2 t^2 (1/R^2 - 1) \xrightarrow{R \gg 1} - 2\sigma_w^2 t^2. \quad (32)$$

Note $V_w'' < 0$: $V_w(\nu)$ has a local *maximum* at ν_t (the variance of the back-integrated w -coordinate *decreases* when you bump ν). $V_z(\nu)$ has a local *minimum*. The two contributions to NLL'' have opposite signs; their truth-sigma sum

$$\text{NLL}''(\nu_t)|_{\text{truth}} = \frac{V_z''(\nu_t)}{\sigma_z^2} + \frac{V_w''(\nu_t)}{\sigma_w^2} \simeq 2t^2(R^2 - 1) + 2t^2(1/R^2 - 1) > 0 \quad (33)$$

is still positive (a true minimum), but dominated by the V_z -term:

$$\boxed{\frac{V_z''(\nu_t)/\sigma_z^2}{|V_w''(\nu_t)|/\sigma_w^2} \simeq R^2.} \quad (34)$$

The z -direction is R^2 times stiffer per unit prior weight than the w -direction.

Physical picture. A trial-frequency error $\delta\nu$ rotates the back-integrated cloud by $\delta\theta = t\delta\nu$ in canonical coordinates. The rotation pumps *velocity into position* with weight σ_w^2/ν_t^2 (large, because the cloud is hot and the orbits are slow) but *position into velocity* with weight $\sigma_z^2\nu_t^2$ (small). Hence V_z is highly sensitive to ν , V_w much less.

2.10 The MLE bias under sigma mismatch

The Newton step around ν_t :

$$\delta\nu_\star \simeq -\frac{\text{NLL}'(\nu_t)}{\text{NLL}''(\nu_t)}. \quad (35)$$

Case A: only σ_w mismatched ($\sigma_{w,\text{fit}} = \sigma_w/r$, $\sigma_{z,\text{fit}} = \sigma_z$).

$$\text{NLL}'(\nu_t) = \frac{V_z'(\nu_t)}{\sigma_z^2} + r^2 \frac{V_w'(\nu_t)}{\sigma_w^2} = \frac{2\sin^2\theta_t}{\nu_t} (r^2 - 1), \quad (36)$$

$$\text{NLL}''(\nu_t) = \frac{V_z''(\nu_t)}{\sigma_z^2} + r^2 \frac{V_w''(\nu_t)}{\sigma_w^2} \simeq 2t^2 R^2 - 2t^2 r^2 \xrightarrow{R \gg r} 2t^2 R^2. \quad (37)$$

$$\boxed{\delta\nu_w \simeq -\frac{(r^2 - 1)\sin^2\theta_t}{\nu_t t^2 R^2}}. \quad (38)$$

Case B: only σ_z mismatched ($\sigma_{z,\text{fit}} = \sigma_z/r$, $\sigma_{w,\text{fit}} = \sigma_w$).

$$\text{NLL}'(\nu_t) = r^2 \frac{V_z'(\nu_t)}{\sigma_z^2} + \frac{V_w'(\nu_t)}{\sigma_w^2} = -\frac{2\sin^2\theta_t}{\nu_t} (r^2 - 1), \quad (39)$$

$$\text{NLL}''(\nu_t) = r^2 \frac{V_z''(\nu_t)}{\sigma_z^2} + \frac{V_w''(\nu_t)}{\sigma_w^2} \simeq 2t^2 R^2 r^2 - 2t^2 \xrightarrow{R^2 r^2 \gg 1} 2t^2 R^2 r^2. \quad (40)$$

The numerator is the same magnitude as Case A — the asymmetry is in the denominator.

$$\boxed{\delta\nu_z \simeq +\frac{(r^2 - 1)\sin^2\theta_t}{\nu_t t^2 R^2 r^2}}. \quad (41)$$

The r^2 asymmetry.

$$\boxed{\frac{|\delta\nu_w|}{|\delta\nu_z|} = r^2}. \quad (42)$$

At $r = 10$: $100\times$ asymmetry. At $r = 100$: $10^4\times$. The σ_w mismatch is overwhelmingly the dominant source of bias.

2.11 One-sentence intuition

Mismatching σ_z amplifies the curvature in the *already-stiff* V_z direction (which dominates NLL'') — the optimum barely shifts. Mismatching σ_w amplifies the *already-soft* V_w direction (which is sub-dominant) — the optimum is pushed far before the small V_w -curvature can stop it.

3 General potential: sech^2 disk + DM

3.1 The actual potential

The simulation uses

$$\Phi(z; h) = \underbrace{\sum_{i \in \{\text{thin, thick, gas}\}} 4\pi G \rho_i h_i^2 \log \cosh(z/h_i)}_{\Phi_{\text{bary}}(z)} + \underbrace{4\pi G \rho_{\text{DM}} h^2 \log \cosh(z/h)}_{\Phi_{\text{DM}}(z; h)}. \quad (43)$$

The baryon component is fixed; the DM component depends on the fitting parameter h .

3.2 No closed-form flow

The equations of motion

$$\dot{z} = w, \quad \dot{w} = -\partial_z \Phi(z; h) \quad (44)$$

are nonlinear and have *no closed-form solution* for any non-trivial sum of $\log \cosh$ terms. The forward map

$$(z_0, w_0) \mapsto (F(z_0, w_0; h, t), G(z_0, w_0; h, t)), \quad (45)$$

where $F = z(t)$ and $G = w(t)$ along the orbit, has to be evaluated numerically. So we cannot write $V_z(h)$ and $V_w(h)$ as elementary functions. We *can*, however, compute their derivatives at $h = h_t$ analytically by linearising — the variational equation.

3.3 The variational equation

Define the orbital perturbation

$$\eta(\tau) \equiv \left. \frac{\partial z(\tau; z_0, w_0, h)}{\partial h} \right|_{h=h_t}, \quad \dot{\eta}(\tau) = \left. \frac{\partial w(\tau; z_0, w_0, h)}{\partial h} \right|_{h=h_t}, \quad (46)$$

i.e. the shift in the orbit at time τ when we bump h slightly, holding (z_0, w_0) fixed. Differentiating (44) with respect to h and evaluating along the true orbit:

$$\boxed{\ddot{\eta}(\tau) + \Phi_{zz}(z(\tau); h_t) \eta(\tau) = -\Phi_{zh}(z(\tau); h_t), \quad \eta(0) = \dot{\eta}(0) = 0,} \quad (47)$$

where $\Phi_{zz} \equiv \partial^2 \Phi / \partial z^2$ and $\Phi_{zh} \equiv \partial^2 \Phi / (\partial z \partial h)$.

This is a *linear* ODE along each orbit — a forced harmonic oscillator with a time-varying “stiffness” $\Phi_{zz}(z(\tau); h_t)$ and a known drive $-\Phi_{zh}(z(\tau); h_t)$. Solvable by the orbit’s Green’s function (Wronskian of the homogeneous tangent flow), which itself has to be evaluated numerically.

3.4 From $\eta(t)$ to $\partial z_0 / \partial h$

Given $(\eta(t), \dot{\eta}(t))$ along each orbit, the implicit function theorem applied to $F(z_0(h), w_0(h); h, t) = z_{\text{obs}}$ and analogous for G — together with the symplectic identity $\det J = 1$ for $J = \partial(F, G) / \partial(z_0, w_0)$ — gives

$$\left. \frac{\partial z_0}{\partial h} \right|_{h_t} = -\eta(t) J_{ww} + \dot{\eta}(t) J_{zw}, \quad (48)$$

$$\left. \frac{\partial w_0}{\partial h} \right|_{h_t} = +\eta(t) J_{wz} - \dot{\eta}(t) J_{zz}, \quad (49)$$

where the tangent-flow Jacobian entries $J_{zz} = \partial F/\partial z_0$, $J_{zw} = \partial F/\partial w_0$, $J_{wz} = \partial G/\partial z_0$, $J_{ww} = \partial G/\partial w_0$ are themselves solutions of the homogeneous variational equation (same ODE as (47) but with zero forcing and unit-vector initial conditions).

3.5 V'_z and V'_w as orbit averages

Then

$$V'_z(h_t) = 2 \left\langle z_0 \frac{\partial z_0}{\partial h} \right\rangle, \quad V'_w(h_t) = 2 \left\langle w_0 \frac{\partial w_0}{\partial h} \right\rangle, \quad (50)$$

and

$$V''_z(h_t) \simeq 2 \left\langle \left(\frac{\partial z_0}{\partial h} \right)^2 \right\rangle, \quad V''_w(h_t) \simeq 2 \left\langle \left(\frac{\partial w_0}{\partial h} \right)^2 \right\rangle. \quad (51)$$

These are exact identities. They cannot be reduced to elementary functions of σ_z, σ_w, h_t — they require numerical orbit integration to evaluate. But the *scaling* is determined by how $\eta(t)$ grows on a typical orbit.

3.6 Why the harmonic scaling persists

Action-angle coordinates: in any 1D Hamiltonian, define

$$J(E; h) = \frac{1}{2\pi} \oint w \, dz, \quad \Omega(J; h) = \frac{dE}{dJ}, \quad (52)$$

so the unperturbed flow is the rigid rotation $\theta(\tau) = \theta_0 + \Omega(J; h_t)\tau$, with J conserved. On a single orbit of frequency $\Omega(J)$, the variational equation (47) has the same linear-secular structure as in the harmonic case: $\eta(\tau)$ grows asymptotically as

$$\eta(t) \sim -\frac{t}{\Omega(J)} w_0 + \text{oscillatory}, \quad (53)$$

$$\dot{\eta}(t) \sim +t\Omega(J) z_0 + \text{oscillatory}, \quad (54)$$

where the prefactors come from the orbit average of Φ_{zh} around the cycle (the “secular” part of the perturbation). This is the *exact same parametric scaling* as the harmonic result, with $\nu \rightarrow \Omega(J)$.

Substituting into (51) and averaging over the action distribution $\mathcal{P}(J)$ of the cloud:

$$V''_z(h_t) \sim 2t^2 \sigma_w^2 \left\langle \frac{1}{\Omega(J)^2} \right\rangle_J, \quad V''_w(h_t) \sim 2t^2 \sigma_z^2 \langle \Omega(J)^2 \rangle_J. \quad (55)$$

Define the orbit-averaged version of R :

$$R_{\text{eff}}^2 \equiv \frac{\sigma_w^2 \langle 1/\Omega^2 \rangle}{\sigma_z^2 \langle \Omega^2 \rangle^{-1}} \approx \frac{\sigma_w^2}{\sigma_z^2 \Omega_{\text{eff}}^2}, \quad \text{with} \quad \Omega_{\text{eff}} \equiv \langle \Omega^2 \rangle^{1/2} \text{ or similar.} \quad (56)$$

Then

$$\frac{V''_z(h_t)/\sigma_z^2}{|V''_w(h_t)|/\sigma_w^2} \simeq R_{\text{eff}}^2. \quad (57)$$

All the harmonic results carry through. Specifically:

- The first-derivative truth-balance (cancellation at $h = h_t$ with truth sigmas) holds exactly — it follows from energy conservation and symplecticity, not from harmonicity.

- The numerator symmetry of the bias ($\propto r^2 - 1$ in either case) is exact — it follows from the form of the Gaussian likelihood alone.
- The stiffness asymmetry $V_z''/\sigma_z^2 \sim R_{\text{eff}}^2 V_w''/\sigma_w^2$ holds whenever the action-distribution is concentrated on orbits with $R_{\text{eff}} \gg 1$.
- The r^2 bias asymmetry (42) between σ_w and σ_z mismatches survives.

3.7 When the harmonic scaling breaks: the $\sigma_z \sim h_{\text{dm}}$ regime

The scaling fails when R_{eff} becomes order unity. This happens precisely when the cloud samples large-amplitude orbits whose frequency $\Omega(J)$ is much smaller than the small-amplitude harmonic limit. Specifically, when σ_z approaches the disk scale-height (or the DM scale h_{dm}), the action distribution moves to low $\Omega(J)$, R_{eff} drops, and the r^2 amplification of the z -stiffness becomes weaker. **This is the regime where you observed σ_z mismatch starting to bias the fit** (the $\sigma_z = 250$ pc test): the previously overwhelmed σ_z direction became a real lever.

3.8 Numerical recipe to verify in the real potential

You can verify the R_{eff}^2 stiffness asymmetry without trusting any approximation, using only the simulation:

1. Fix h_t and generate the data (one cloud).
2. Pick a fine grid $\{h_k\}$ around h_t . For each h_k , back-integrate the data (no fitting) and compute $V_z(h_k)$ and $V_w(h_k)$.
3. Fit a parabola in h to each curve; read off $V_z''(h_t)$ and $V_w''(h_t)$.
4. Compute the per- σ^2 ratio

$$\rho \equiv \frac{V_z''(h_t)/\sigma_z^2}{|V_w''(h_t)|/\sigma_w^2}. \quad (58)$$

5. Compute R_{eff}^2 directly: from the data cloud, evaluate the energy of each star, find its $\Omega(J; h_t)$ (a 1D quadrature in the potential), and form $R_{\text{eff}}^2 = \sigma_w^2/(\sigma_z^2 \langle \Omega^2 \rangle)$.
6. If $\rho \approx R_{\text{eff}}^2$, the harmonic scaling is confirmed in the real potential.

4 Summary

1. The MLE balance (15) is exact in any potential.
2. In the harmonic case, $V_z(\nu)$ and $V_w(\nu)$ are computed in closed form (7). First derivatives (22) have equal magnitude and opposite sign — the truth-sigmas balance them. Second derivatives satisfy the stiffness asymmetry (34): V_z is R^2 -times stiffer per unit prior weight than V_w .
3. Mismatch by factor r produces a numerator (force) that is symmetric between σ_z and σ_w ($\propto r^2 - 1$), but a denominator (curvature) amplified by r^2 when σ_z is mismatched (stiff direction) and barely changed when σ_w is mismatched (soft direction). Hence the r^2 bias asymmetry (42).

4. In the real sech^2 +DM potential, the same structure follows from the variational equation (47), with $\nu \rightarrow \Omega(J)$ averaged over the action distribution. The harmonic R becomes an orbit-averaged R_{eff} , but the r^2 bias asymmetry survives whenever the cloud sits on orbits with $R_{\text{eff}} \gg 1$.
5. The asymmetry breaks down when σ_z becomes large compared to the disk scale-height (the cloud reaches low- Ω orbits) — exactly the regime where your $\sigma_z = 250$ pc test showed bias re-emerging.